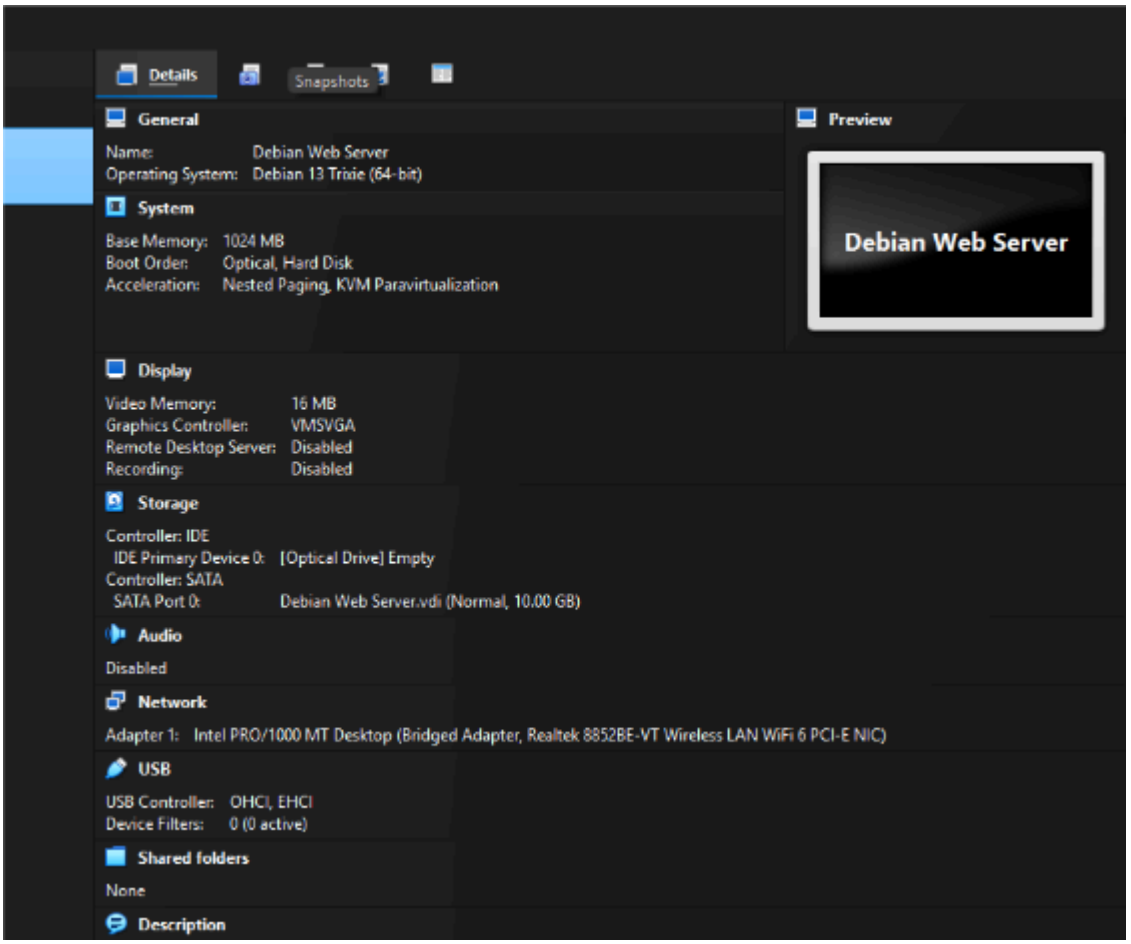
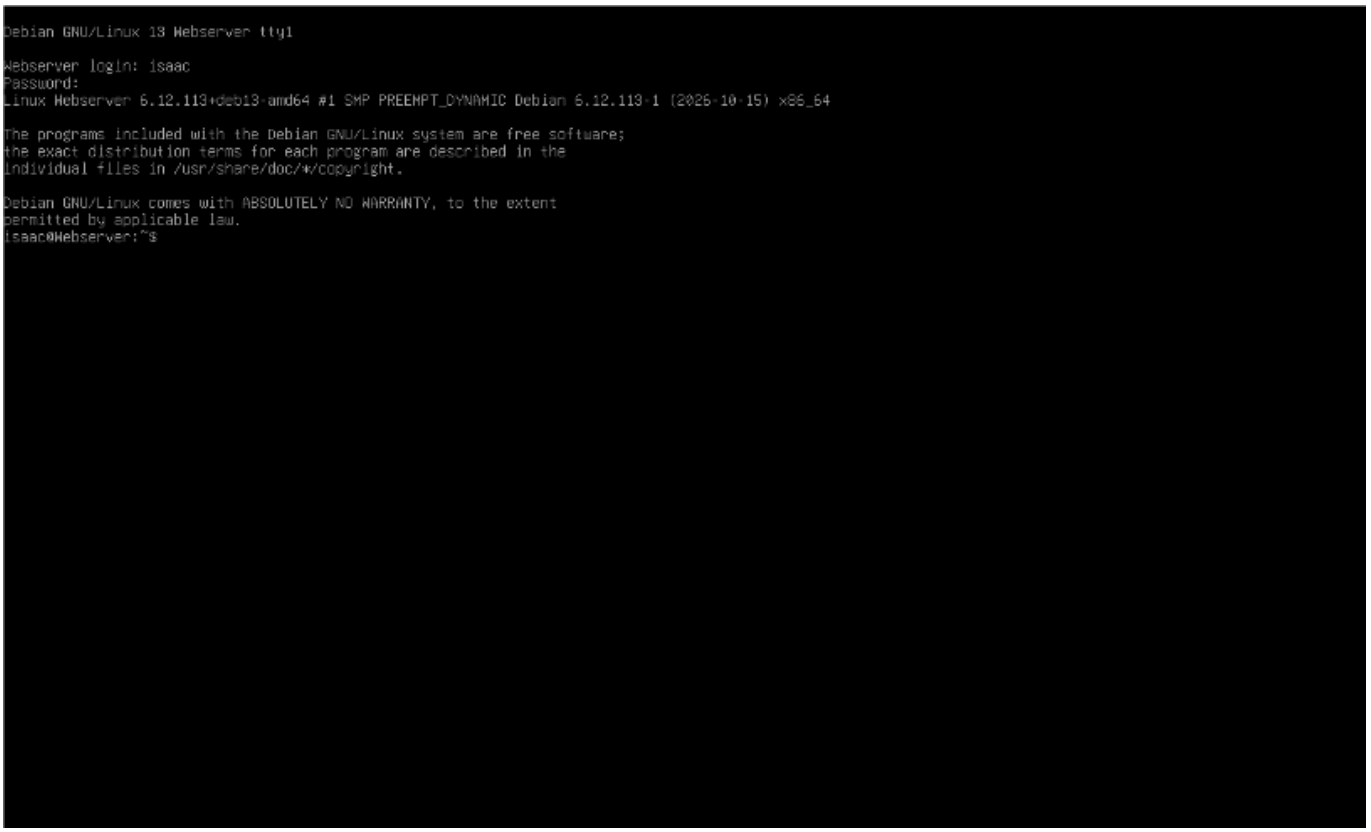
Server Hardware specifications



Debian Login screen



IP address debian server



How to work with the Firewall

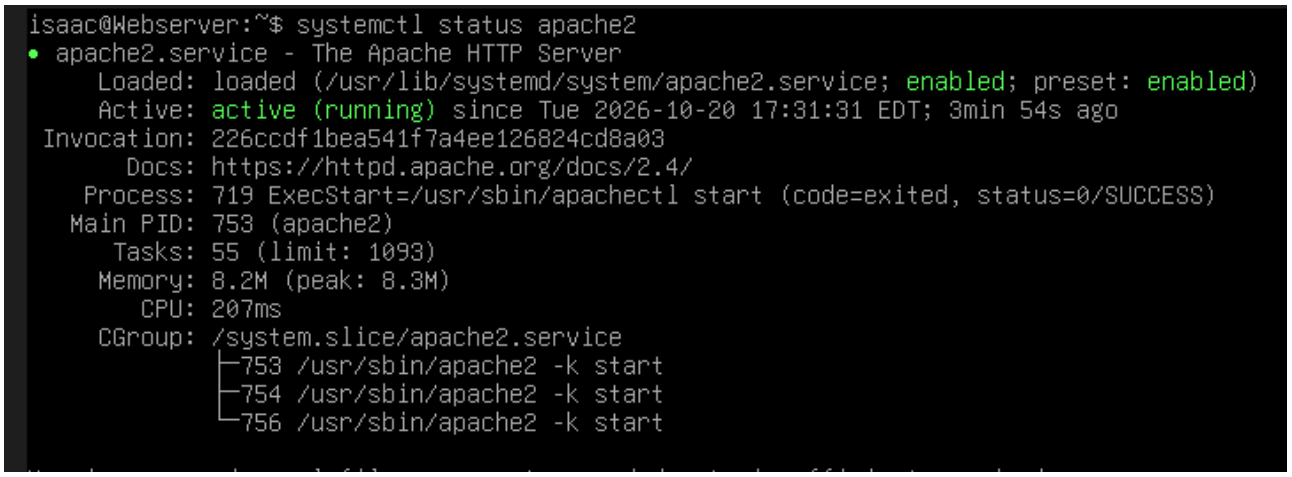
- **Description**: In Debian, the firewall is managed with **ufw** (Uncomplicated Firewall). It controls which network traffic is allowed in or out of the system by turning the firewall on or off and adding or removing rules.
- **Formula/Syntax**: `sudo ufw` + `action` + `argument`
- `sudo ufw` is the command (run with admin rights)
- `action` is what to do, such as `status`, `enable`, `disable`, or `allow`
- `argument` is what the action applies to, such as an application profile like `'WWW Full'`
- **Examples**:
- How do you check if the Firewall is running?

- By using the command `sudo ufw status`, which shows whether the firewall is active or inactive and lists the rules. You can also use `systemctl status ufw` to check the ufw service.
- How do you disable the Firewall?
- By using the command `sudo ufw disable`, which turns the firewall off so no rules are enforced.
- How do you add Apache to the Firewall?
- By using the command `sudo ufw allow 'WWW Full'`, which allows web traffic on ports 80 (HTTP) and 443 (HTTPS).

What different commands to use to work with Apache2

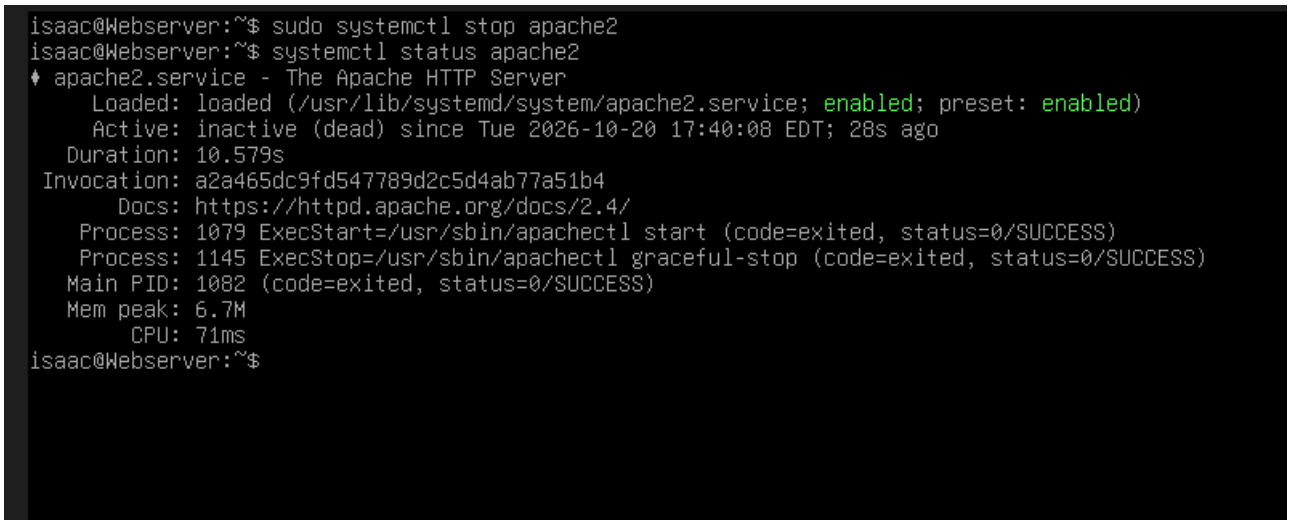
1. What is the command you use to check if Apache is running?

- The command is: `systemctl status apache2`
- What it does: shows whether the Apache service is `active (running)` or `inactive (dead)`, along with its most recent log lines.



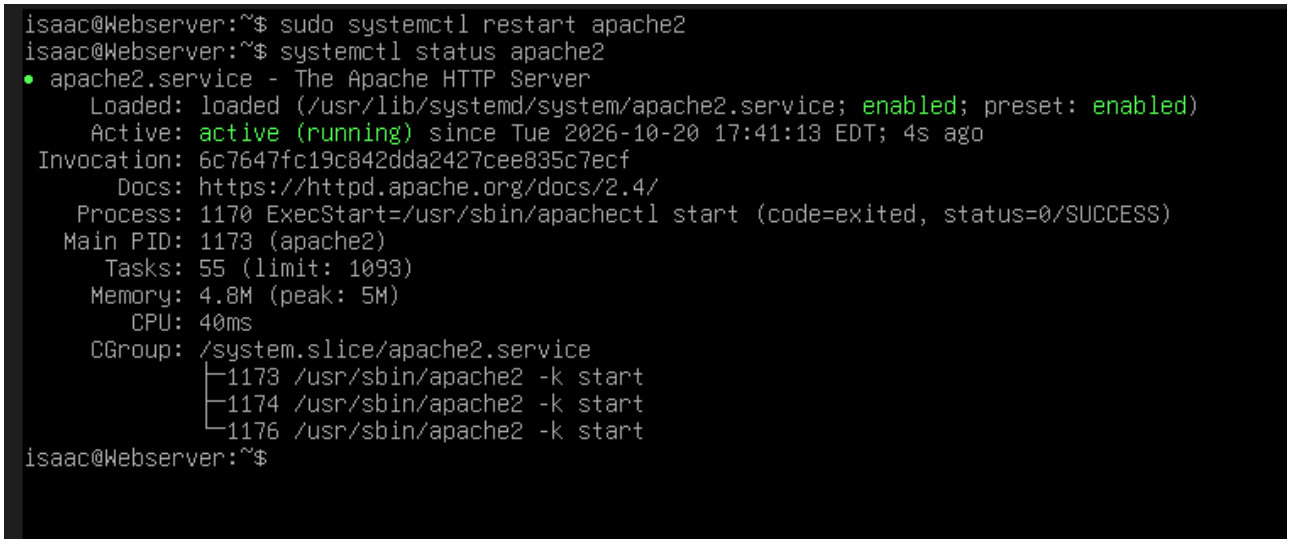
2. What is the command you use to stop Apache?

- The command is: `sudo systemctl stop apache2`
- What it does: shuts down the Apache service so the website no longer loads. It prints nothing when it works.



3. What is the command you use to restart Apache?

- The command is: `sudo systemctl restart apache2`
- What it does: stops Apache and starts it again. This applies configuration changes or turns Apache back on after it was stopped.



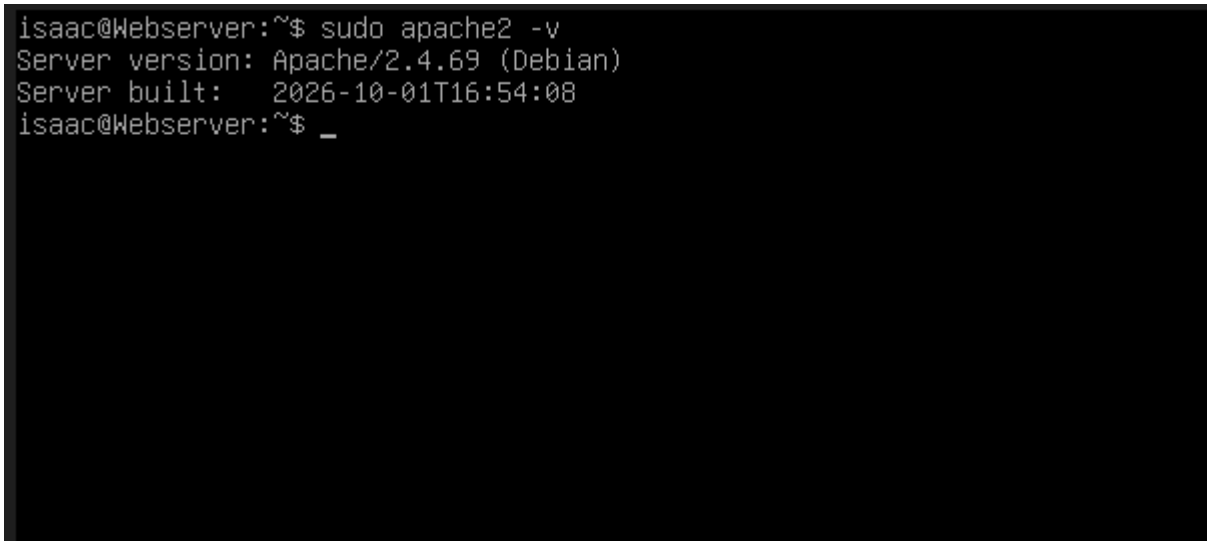
4. What is the command used to test Apache configuration?

- The command is: `sudo apache2ctl configtest`
- What it does: checks the Apache configuration files for errors without restarting the server. `Syntax OK` means the configuration is valid.



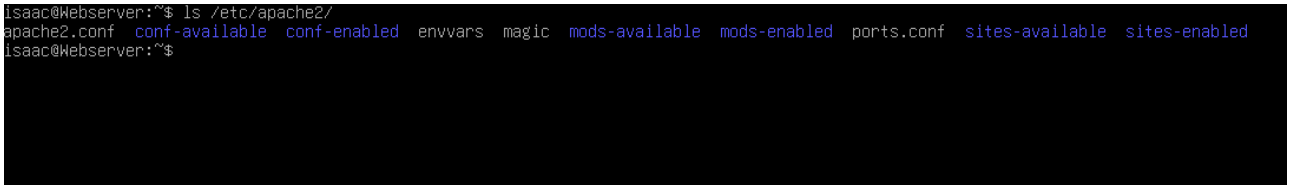
5. What is the command used to check the installed version of Apache?

- The command is: `sudo apache2 -v`
- What it does: prints the installed Apache version number and the date it was built.



6. What are some common configuration files for Apache?

- The command is: `ls /etc/apache2/`
- What it does: lists the files and folders in Apache's configuration directory. Common ones are `apache2.conf` (main configuration), `ports.conf` (ports Apache listens on), `sites-available` (virtual hosts), `mods-available` (modules) and `conf-available` (extra settings).



7. Where does Apache store logs?

- The command is: `sudo ls /var/log/apache2/`
- What it does: lists the files in the Apache log directory, `/var/log/apache2/`. It contains `access.log` (requests to the server) and `error.log` (errors). `sudo` is needed because the folder is restricted.



8. What are some basic commands we can use to review logs?

- The command is: `sudo tail -f /var/log/apache2/access.log`

- `tail -f` follows the log live, showing new entries as they are written.

